Ajay Venkatesh

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Education

New York University, New York, NY — *Aug 2024 – May 2026*

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: System Design, Cloud Computing, Machine Learning, Big Data, Advanced Java, Data Structures

Experience

- **CampusMesh** — *Jan 2026 – May 2026*
  Software Engineering Intern — New York, US
  - Architected a **serverless backend** on AWS using **Serverless Framework**, **Lambda (Node.js 20)**, API Gateway, RDS/MySQL, DynamoDB, S3, and CloudFront; deployed across **4 environments** with SSM configs, KMS keys, and **least-privilege IAM**.
  - Integrated **AWS Chime SDK** for in-space audio/video calls with idempotent meeting creation and a scheduled Lambda sweeper; modeled high-throughput chat in DynamoDB with partition+sort keys to eliminate hot partitions.
  - Built a **JWT custom authorizer** with **AES-encrypted** user IDs and esbuild-bundled Lambdas to cut cold starts; designed a layered handler → service → repository → model architecture across **20+ endpoints** for the Study Space service.
- **Amazon** — *May 2025 – Aug 2025*
  Software Development Engineer Intern — Seattle, US
  - Designed a secure UI Console architecture using **Midway SSO, OAuth, and HELIS** to enforce least-privilege **role-based access** (**4+ roles**); presented the solution to the team and received design approval.
  - Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, and IAM; built a **multi-stage, multi-region CI/CD pipeline** with automated promotion and rollback controls.
  - Built a **React 18** frontend with optimized API integration, advanced filtering, and client-side caching, reducing page load time by **40%**.
  - Engineered backend services in **Java** (Coral, Smithy, Dagger DI, CBOR); applied **low-level design patterns**, cutting backend execution time by **35%** with unit/integration testing via **Mockito**.
- **New York University (Novel AI Technologies, Inc.)** — *Aug 2024 – Present*
  Software Developer — New York, US
  - Architected an **NSF-funded ($100K)** AI-powered health app for **iOS** and **Android** processing real-time wearable data; deployed cloud-hosted AI models for health metric analysis and food image processing.
  - Engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with **concurrency control and locking**; processed **250K+** records and powered **AWS QuickSight** dashboards.
  - Developed an insurance analytics platform (**React, Node.js, REST APIs**) deployed via **Docker** on **EC2**; implemented Stripe billing, RBAC, and row-level security, generating **$50K+** revenue across **6 clients**.
  - Streamlined underwriting via a broker portal integrated with insurer platforms; implemented **document/media processing** and **WebSocket-based real-time communication**.
- **PricewaterhouseCoopers (PwC)** — *Aug 2021 – Aug 2024*
  Senior Software Engineer — Bengaluru, India
  - Designed and implemented a **cloud-native distributed backend** on **Microsoft Azure** for a health platform, integrating **Microsoft Dynamics CRM** and event-driven Azure services for scalability and fault tolerance over **SQL Server**.
  - Modernized a legacy onboarding system by building a scalable **B2B platform** with modular services; integrated data pipelines, email event tracking, and BI reporting, improving automation by **70%**.
  - Deployed an **NLP-based OCR neural model** on Azure to automate structured data extraction, reducing processing time by **75%**.
  - Optimized data ingestion with **parallel processing** on **100K-row** Excel feeds, cutting execution from **5 hours to 1 hour**; mentored engineers on **software design principles**, reducing production bugs by **25%**.

Projects

- **InviLite** | *AWS, SageMaker, Bedrock, FAISS, Python*
  - Architected a cloud-native data pipeline (Kinesis, Lambda, S3, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed ML pipelines.
  - Built a **RAG system** (SVM + FAISS + **Bedrock LLM**), cutting incident resolution from **1–2 hours to minutes** and lifting root-cause classification accuracy from **56% to 92%**.
- **Flight Delay Prediction** | *PySpark, HDFS, Random Forest, Gradient Boosted Tree*
  - Built a scalable **PySpark ML pipeline** on HDFS to process and engineer features from over **7M flight records** using SQL-based aggregation and anomaly detection.
  - Trained and optimized **Random Forest and Gradient Boosted Tree** models reaching **85%+** accuracy and converted outputs into actionable insights for data-driven scheduling.

Skills

- **Programming Languages:** C#, Python, Java, C++, JavaScript, TypeScript, SQL
- **Cloud (Azure):** Azure AI Services, Azure AI Foundry, Cognitive Services, Logic Apps, Function Apps, Storage Blobs, Data Lakes, Azure DevOps, Azure Logic Apps
- **Cloud (AWS):** CDK, SDK, EC2, S3, Lambda, IAM, SageMaker, Bedrock, API Gateway, DynamoDB
- **Databases:** SQL Server, Dataverse, PostgreSQL, MySQL, MongoDB, DynamoDB
- **Microsoft Stack:** .NET, Power Platform, Power Automate, Power BI, Microsoft Copilot Studio, Microsoft Dynamics CRM
- **DevOps & Tools:** Docker, Kubernetes, CI/CD Pipelines, PowerShell, Git, Visual Studio, Postman, AMQP, HL7, FHIR, DICOM, PACS, LIS, RIS
- **AI/ML:** RAG, MLOps, AI Solution Deployment, Responsible AI, Model Validation & Monitoring

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques *(IEEE)*

An Aquila-optimized SVM Classifier for Diabetes Prediction *(IEEE)*

Certifications

AWS Certified Cloud Practitioner | Microsoft Certified: Azure Data Scientist Associate | Microsoft Certified: Azure Data Fundamentals